

RoboJev: Confidence-Gated Structured Decision Making for Long-Horizon Embodied Agents

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Preprint, 2026

Abstract

Long-horizon embodied agents repeatedly face closed-form questions during execution: Did the grasp hold? Did the skill succeed? Should the controller retry, replan, or escalate? Current stacks often answer these questions by invoking an LLM or VLM at every step, even when the answer space is tiny. We present RoboJev, a three-layer architecture that separates open-ended reasoning from high-frequency structured decisions. A Reasoning layer (LLM/VLM) writes or revises skill plans only when needed. A Decision layer returns typed judgments with confidence in the style of System-One models such as Jev. An Action layer executes skills. Confidence-gated escalation routes only uncertain cases back to Reasoning. In a symbolic kitchen task with MuJoCo display rendering, RoboJev completes clean, slip, ambiguous-sensor, and early door-close episodes while cutting reasoner calls versus a replan-every-step baseline.

1. Introduction

Embodied agents that execute multi-step skills spend most of their time not inventing new language, but judging short, discrete outcomes. Grasp success, skill completion, failure type, and recovery choice are exactly the kind of high-frequency, schema-constrained decisions that generative models are expensive at answering. Recent System-One models argue that automation needs fast, typed judgments rather than another paragraph of free text.

RoboJev applies that idea to embodied control. The claim is architectural: if a question has a small answer space, it belongs in a Decision layer with confidence; only low-confidence or open-ended cases should escalate to an LLM. The contribution of this preprint is a minimal, reproducible loop that makes the claim concrete without requiring real-robot hardware for control.

2. Related Work

Hierarchical robot stacks commonly separate planning from control (task-and-motion planning, skill libraries, VLA policies). LLM-based agents increasingly act as the planner, but often also as the step-level critic. Selective prediction and confidence gating have long been used to defer uncertain decisions. RoboJev sits between these lines: it keeps LLMs for open planning, while assigning closed judgments to a structured System-One interface inspired by Jev.

3. Method

RoboJev factors an episode into Reasoning, Decision, and Action. The environment exposes a symbolic belief: robot location, gripper contents, object locations, and container openness. Skills are pick, move, open, place, and close. Perception for control is state text; RGB frames are rendered only for display.

3.1 Reasoning

The Reasoning layer proposes an ordered skill plan from the current belief. In our implementation it is an LLM with a constrained JSON schema; a scripted planner is used as a fallback and for ablations. Reasoning is called at episode start, on explicit replan, and when Decision escalates.

3.2 Decision

After each skill, Decision evaluates four questions in one shot: (1) `grasp_ok` (noul), (2) `skill_done` (noul), (3) failure type (choice among none/slip/miss/blocked/uncertain), and (4) recovery (continue/retry/replan/abort). Answers include calibrated confidence. If recovery confidence is below 0.6, or `skill_done` falls in (0.35, 0.65), the controller escalates to Reasoning with a clarified belief.

3.3 Action

Action executes the selected skill in the symbolic world and returns a sensor reading used by Decision. MuJoCo mirrors the same state for screenshots; the agent never conditions on pixels.

4. Experiments

Task: put an apple into a fridge and close the door. We evaluate four scripted episodes on one NVIDIA H800, with Decision as a local System-One stand-in matching Jev’s answer schema, and Reasoning via a hosted chat model. The baseline replans with the reasoner after every skill.

Episode	RoboJev reas.	Baseline reas.	Decisions	Outcome
Clean	1	6	5	success
Slip	1	7	6	retry
Ambiguous	2	6	5	escalate
Lid closes	2	8	7	replan

Clean runs need a single plan. Slip is recovered by a high-confidence local retry. Ambiguous sensors escalate once. An early door close triggers replan rather than blind retries. Across episodes, RoboJev uses far fewer reasoner calls than the baseline while finishing the task.

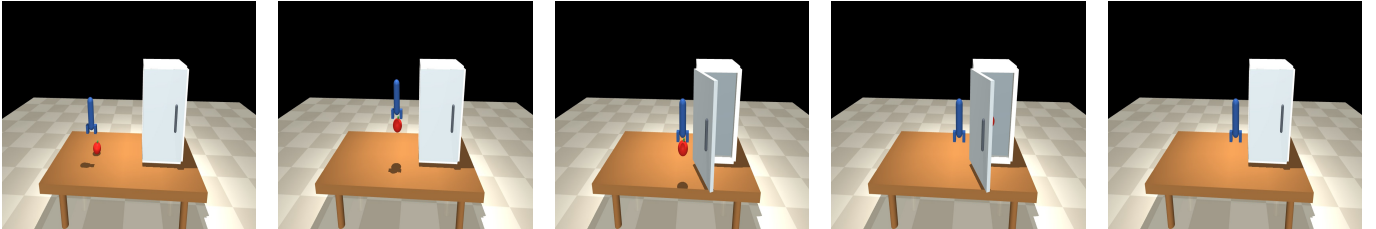


Figure 1. Clean execution. Display frames from MuJoCo; control uses state only.

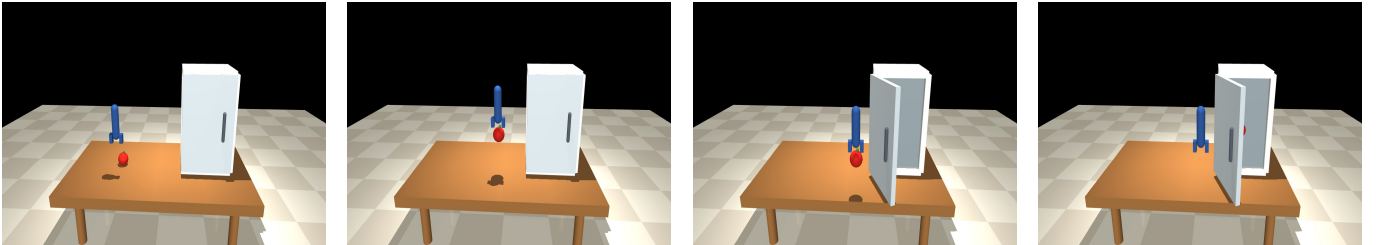


Figure 2. Left to right: slip then retry; ambiguous open (escalate); blocked place before replan.

5. Limitations and Future Work

This preprint is intentionally minimal. The world is symbolic; Decision is a schema-compatible local model unless a hosted System-One API is provided; and we do not claim EDBench-scale evaluation, real-robot transfer, or learned calibration. Failure modes include over-eager escalation, brittle LLM plan schemas, and display scenes that overstate geometric fidelity.

Future work includes replacing the local Decision stand-in with a trained System-One model on robot traces, evaluating on standard embodied benchmarks, closing the loop with real sensors, and studying when confidence is trustworthy enough to suppress LLM calls safely.

6. Conclusion

RoboJev is a simple recipe: let LLMs reason when the problem is open, and let structured judgments handle the high-frequency closed questions that dominate long-horizon execution. Even a minimal kitchen loop shows that confidence-gated escalation can preserve competence while cutting reasoner calls.

Acknowledgments

This preprint accompanies the project page at <https://robojev.github.io>. Code is released at <https://github.com/robojev/robojev>.

References

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